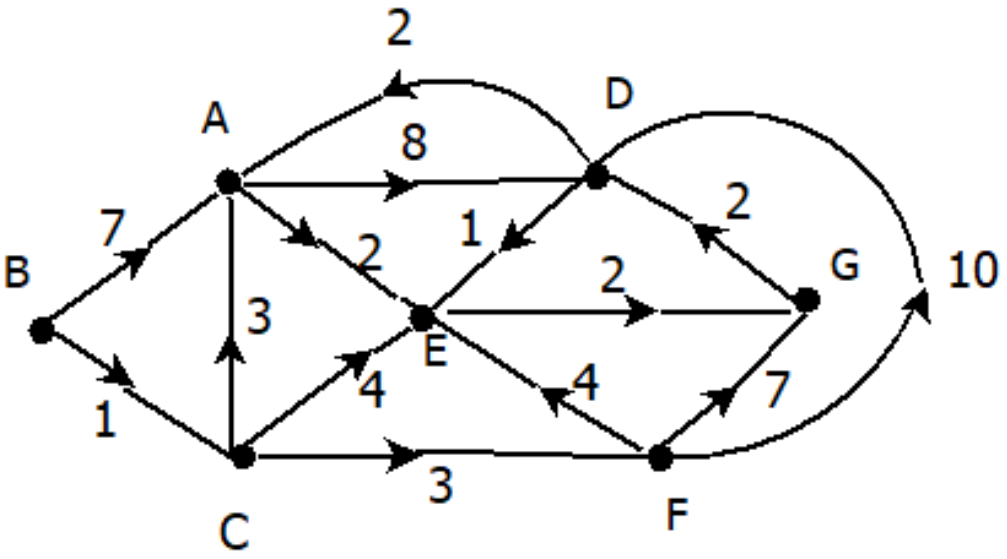
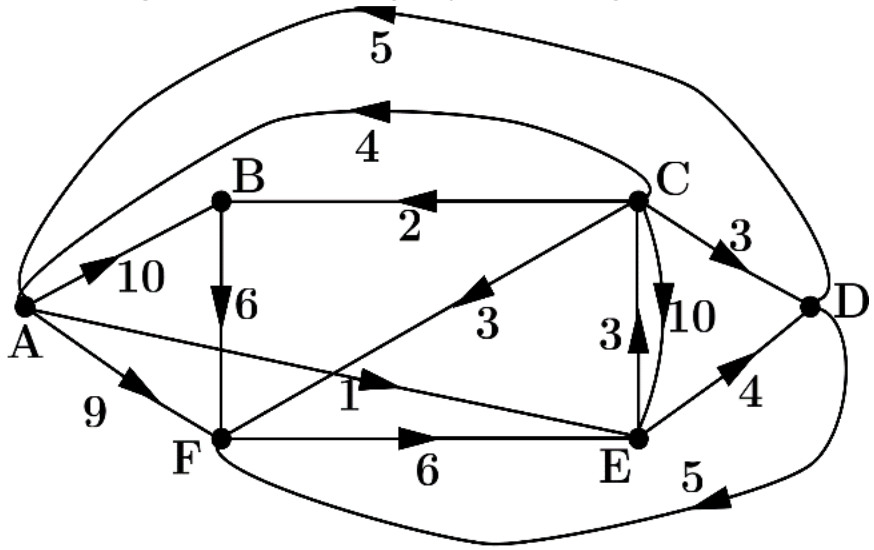
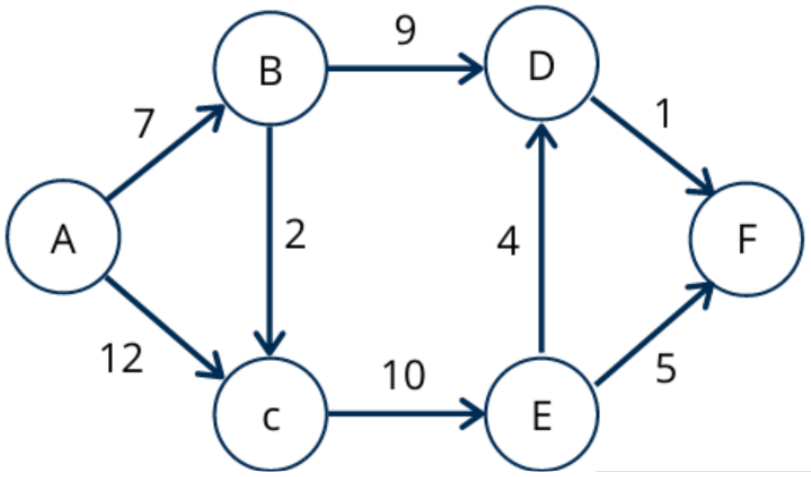
MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

Assignment

Engineering Mathematics III(MAT_2155/MAT_2126)

Number of questions:10 **Last date to submit: 30th October 2024**

S. No	Question	CO
1.	Find the 27th and 25th permutation of 5 marks 0,1,2,3,4 in both (i) Lexicographical (ii)Reverse Lexicographical ordering and (iii) Fike's ordering.	1
2.	Find the 35th permutation of 5 marks 1,2,3,4,5 in both (i) Lexicographical (ii)Reverse Lexicographical ordering and (iii) Fike's ordering.	1
3.	Find the 17th permutation of 4 marks $\alpha, \beta, \gamma, \omega$ in both (i) Lexicographical (ii)Reverse Lexicographical ordering and (iii) Fike's ordering.	1
4.	Find the shortest path from B to all other vertices for the network given in the diagram below using Dijkstra's Algorithm. 	1

5.	Find the shortest path from A to all other vertices for the network given in the diagram below using Dijkstra's Algorithm. 	**1**
6.	Find distance matrix of the following graph. Using Dijkstra algorithm find shortest path from the source vertex A to all other vertices. 	**1**
7.	Let $E(x_1, x_2, x_3) = \overline{(x_1 \wedge x_2)} \vee (\overline{x_2} \wedge x_3)$ be a Boolean expression over the two values Boolean Algebra. Express E in conjunctive normal form and disjunctive normal form.	**4**
8.	Let $E(x_1, x_2, x_3) = \overline{(\overline{(x_1 \wedge x_2)} \vee (\overline{x_2} \wedge x_3)}$ be a Boolean Expression over the two-valued Boolean Algebra. Write $E(x_1, x_2, x_3)$ in both DNF and CNF.	**4**
9.	Let $E(x_1, x_2, x_3) = (\overline{x_1} \wedge x_2 \wedge \overline{x_3}) \vee (x_1 \wedge \overline{x_2} \wedge x_3) \vee (x_1 \wedge x_2 \wedge x_3)$ be a Boolean expression over the two valued Boolean algebra $\{0, 1\}$. Express $E(x_1, x_2, x_3)$ as a both disjunctive and conjunctive normal form.	**4**
10.	Let $E(x_1, x_2, x_3) = (\overline{x_1} \wedge x_2 \wedge \overline{x_3}) \vee (x_1 \wedge \overline{x_2}) \vee (x_1 \wedge x_3)$ be a Boolean expression over the two valued Boolean algebra $\{0, 1\}$. Express $E(x_1, x_2, x_3)$ as a both disjunctive and conjunctive normal form.	**4**